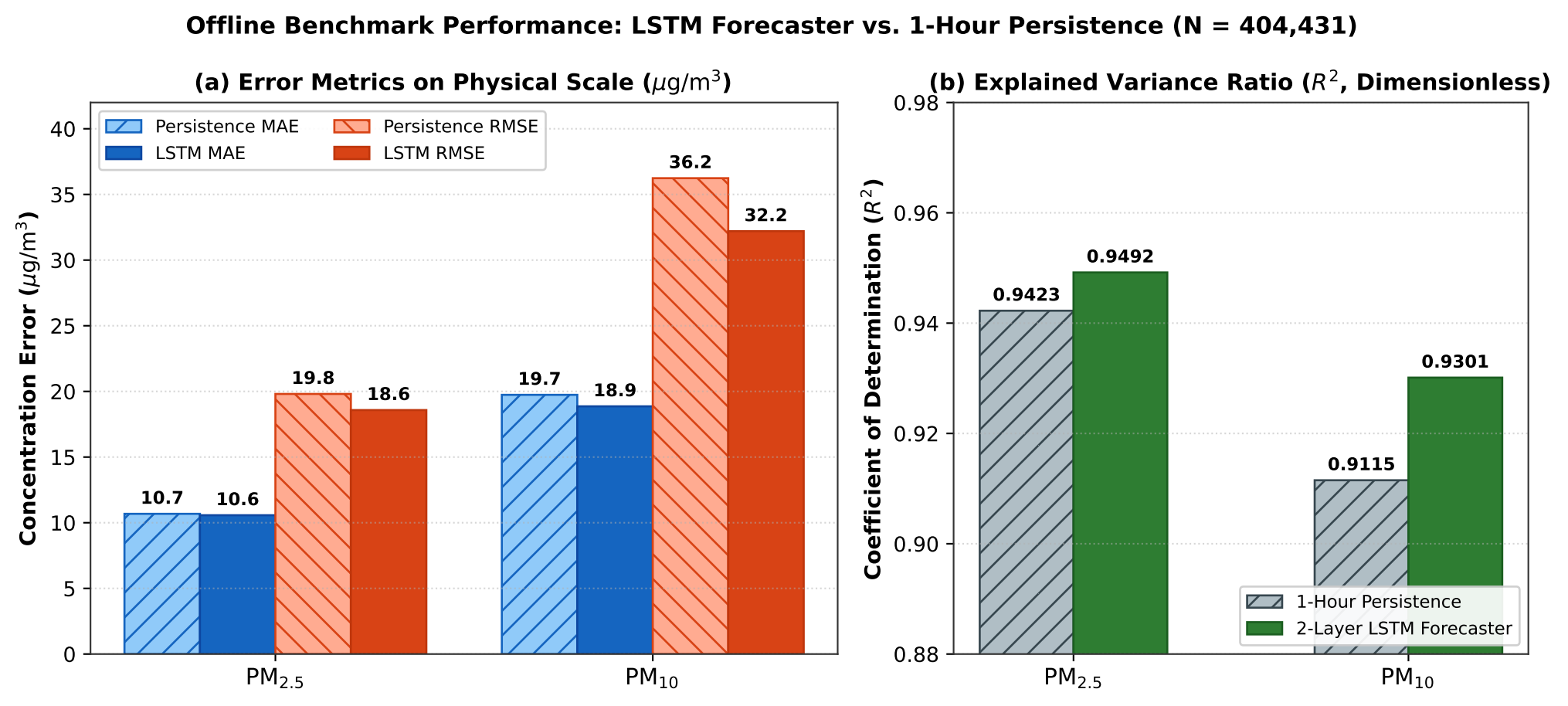
Offline Benchmark Performance: LSTM Forecaster vs. 1-Hour Persistence (N = 404,431)
(a) Error Metrics on Physical Scale (μg/m³)
Concentration Error (μg/m³)
Persistence MAE
LSTM MAE
Persistence RMSE
LSTM RMSE
10.7
10.6
19.8
18.6
19.7
18.9
36.2
32.2
PM2.5
PM10
(b) Explained Variance Ratio (R², Dimensionless)
Coefficient of Determination (R²)
0.9423
0.9492
0.9115
0.9301
1-Hour Persistence
2-Layer LSTM Forecaster